

TE-09 CHEMICAL / HAZMAT EXPOSURE

Toxicology & Environmental | EMT - AEMT - Paramedic

Responder safety comes first. Do not enter a hot zone, vapor cloud, confined space, or oxygen-deficient atmosphere without the required training, atmospheric monitoring, chemical PPE, respiratory protection, and HazMat coordination.

QUICK FLOW

1. ZONES + PPE

Hot, warm, cold zones; wind, terrain, vapor, confined-space risk.

2. SAFE RESCUE ONLY

Removal by properly trained and protected personnel.

3. DECONTAMINATE

Remove clothing; brush dry agents; blot liquids; irrigate as indicated.

4. ABCs + THREATS

Airway edema, bronchospasm, failure, shock, seizure, dysrhythmia.

5. IDENTIFY

SDS/ERG, label, placard, route, concentration, duration, symptoms.

6. POISON HELP

1-800-222-1222; agent guidance, antidote, destination.

GENERAL DECONTAMINATION

- Keep contaminated patients, clothing, equipment, packaging, and runoff out of the clean treatment area, ambulance, and receiving facility.
- Remove and bag clothing and jewelry. Brush visible dry material before irrigation; blot excess liquid. Follow HazMat direction for water-reactive agents and runoff.
- For most liquid eye or skin exposures, begin copious tepid-water or saline irrigation immediately. Remove contact lenses when easily possible without delaying irrigation.
- Obtain the product name, label, placard, SDS, shipping papers, exposure route, duration, time, and symptoms without touching contaminated materials or re-entering the hazard.

PROVIDER-LEVEL ACTIONS

EMT	Scene control, PPE limits, clothing removal, dry-agent brushing, irrigation, airway/ventilation, oxygen, authorized bronchospasm care, product information, and early Poison Help contact.
AEMT	All EMT care plus IV/IO access in an unexposed extremity when feasible, fluids, authorized medications, monitoring, and preparation for rapid deterioration.
PARAMEDIC	All prior care plus advanced airway/ventilation, ECG, 12-lead, seizure, dysrhythmia and shock care. Use specialty antidotes only through approved standing orders or direct Medical Control.

GENERAL TREATMENT + TRANSPORT

- Support airway, suction, oxygenation and ventilation; treat bronchospasm, shock, seizure, dysrhythmia, burns, temperature injury, and pain using the appropriate linked protocol.
- Do not rely on pulse oximetry alone when cellular asphyxiation, carbon monoxide, cyanide, hydrogen sulfide, or methemoglobinemia is possible.
- Call Poison Help and Medical Control early for severe, unknown, pediatric, pregnant, intentional, ingestion, delayed-risk, or antidote-requiring exposure.
- Notify the receiving facility before arrival with agent, route, symptoms, treatments, Poison Help recommendations, and decontamination status.

QUICK CARD 1: CHLORINE / CHLORAMINE GAS

Common sources: pool chemicals; industrial chlorine; bleach mixed with acid or ammonia; liquid chlorine products

PRIMARY HAZARDS

Major threat: corrosive injury to moist tissue, bronchospasm, airway injury, hypoxemia, and delayed pulmonary edema. Liquid-soaked clothing may continue to off-gas.

RECOGNITION

- Burning eyes, tearing, throat irritation, cough, chest tightness, wheeze, rales, stridor, dyspnea, hypoxemia, or pulmonary edema.
- Skin pain, inflammation, blistering, or frostbite after exposure to liquefied chlorine.
- Symptoms may worsen after removal from exposure; repeat lung assessment and oxygenation/ventilation monitoring.

FIELD PRIORITIES

- Protected removal to fresh air; remove clothing and jewelry; bag contaminated items; wash skin with mild soap and water; irrigate eyes.
- Support ventilation, administer oxygen as indicated, treat bronchospasm, and prepare for airway or CPAP/BVM needs according to patient condition.
- Do not routinely neutralize. Nebulized sodium bicarbonate requires specific Claiborne adoption, approved preparation, and protocol or Medical Control order.

CRITICAL ALERT

Never mix chemicals to identify them. Mixing bleach with acid or ammonia can generate additional toxic gas.

QUICK CARD 2: AMMONIA

Common sources: fertilizer; agricultural operations; industrial refrigeration; concentrated cleaners

PRIMARY HAZARDS

Major threat: highly water-soluble alkaline injury causing severe eye and upper-airway damage, laryngeal edema, bronchospasm, burns, hypotension, and pulmonary edema.

RECOGNITION

- Severe eye pain or conjunctivitis, nasal/throat burning, cough, hoarseness, stridor, wheeze, dyspnea, or progressive airway swelling.
- Skin burns or frostbite from liquefied product; GI injury may occur after ingestion.
- Airway findings can progress rapidly after a high-concentration exposure.

FIELD PRIORITIES

- Protected rescue; remove clothing and jewelry; brush/blot material; wash skin; begin continuous eye irrigation.
- Support oxygenation and ventilation, treat bronchospasm, monitor ECG and perfusion, and prepare early for a progressively difficult airway.
- Do not delay transport for persistent eye pain, airway symptoms, abnormal lung findings, burns, hypotension, or significant enclosed-space exposure.

CRITICAL ALERT

Early airway planning matters. Progressive laryngeal edema may make later airway management substantially more difficult.

QUICK CARD 3: ORGANOPHOSPHATE / CARBAMATE

Common sources: agricultural pesticides; insecticides; contaminated clothing, equipment, runoff, or intentional ingestion

PRIMARY HAZARDS

Major threat: cholinergic respiratory failure from bronchospasm, profuse pulmonary secretions, muscle weakness, seizure, and secondary responder contamination.

RECOGNITION

- Miosis, sweating, salivation, lacrimation, vomiting, diarrhea, urination, abdominal cramps, fasciculations, weakness, confusion, bradycardia or tachycardia.
- Bronchospasm, copious secretions, hypoxemia, respiratory fatigue, seizure, or coma indicate severe toxicity.
- Contaminated skin, hair, clothing, emesis, and secretions may expose rescuers.

FIELD PRIORITIES

- Use chemical PPE; remove clothing; bag items; brush dry material; wash skin and hair with soap and water; protect the airway during decontamination.
- Aggressive suction, oxygenation and ventilation are immediate priorities.
- Use TE-08 WMD / Nerve Agent for atropine, pralidoxime, autoinjectors, seizure care, and CHEMPACK procedures.

CRITICAL ALERT

Atropine endpoint: improving bronchial secretions and ventilation - not pupil size, a fixed heart rate, or a predetermined maximum.

QUICK CARD 4: HYDROGEN SULFIDE

Common sources: sewers; septic systems; manure pits; petroleum operations; decaying organic matter; low or confined spaces

PRIMARY HAZARDS

Major threat: rapid incapacitation, cellular asphyxiation, respiratory arrest, seizure, dysrhythmia, circulatory collapse, and multiple-rescuer fatalities.

RECOGNITION

- Rotten-egg odor is unreliable because olfactory fatigue occurs. Absence of odor does not indicate safety.
- Eye irritation, cough, dyspnea, pulmonary edema, headache, dizziness, confusion, seizure, coma, sudden collapse, or respiratory arrest.
- The gas is heavier than air and can concentrate in pits, tanks, sewers, and other low spaces.

FIELD PRIORITIES

- Only trained personnel with atmospheric monitoring and appropriate respiratory protection may enter for rescue.
- Provide immediate high-concentration oxygen and ventilation, cardiac monitoring, seizure care, and shock management.
- Contact Poison Help and Medical Control. Nitrite therapy is used only if specifically approved and clinically appropriate.

CRITICAL ALERT

Do not become the next patient. An unprotected confined-space rescue can produce multiple fatalities within seconds.

QUICK CARD 5: HYDROFLUORIC ACID / HYDROGEN FLUORIDE

Common sources: glass etching; rust removal; metal or wheel cleaners; industrial and laboratory products

PRIMARY HAZARDS

Major threat: deep tissue fluoride injury with hypocalcemia, hypomagnesemia, hyperkalemia, prolonged QT, ventricular dysrhythmia, shock, and death - sometimes from a small-appearing burn.

RECOGNITION

- Severe or delayed pain, often out of proportion to visible injury; whitening, erythema, or deep tissue injury may be subtle.
- Palpitations, weakness, hypotension, prolonged QT, ventricular ectopy, dysrhythmia, seizure, or collapse suggest systemic toxicity.
- Eye, inhalation, ingestion, hand, face, genital, circumferential, large-area, or symptomatic exposures are high risk.

FIELD PRIORITIES

- Use chemical PPE; remove contaminated items; brush dry material; irrigate immediately and copiously.
- Establish IV/IO access in an unexposed extremity when feasible and begin continuous ECG monitoring.
- After irrigation, calcium gluconate treatment follows only the finalized Claiborne HF kit/protocol and Poison Help or Medical Control direction.

CRITICAL ALERT

Do not judge severity by burn size alone. Persistent pain or ECG change after HF exposure requires urgent specialty care.

QUICK CARD 6: UNKNOWN CHEMICAL / CORROSIVE

Common sources: unlabeled products; spills; laboratories; transportation incidents; household or industrial acids/alkalis

PRIMARY HAZARDS

Major threat: unidentified responder hazard, ongoing contamination, airway or eye injury, caustic burns, aspiration, delayed pulmonary toxicity, or an agent that reacts dangerously with water.

RECOGNITION

- Use placards, labels, SDS, container shape, ERG information, witnesses, setting, and toxidrome without touching or transporting contaminated containers.
- Assess airway, lung sounds, eyes, skin, mental status, secretions, perfusion, ECG, and evidence of ingestion or intentional exposure.
- A normal initial examination does not exclude delayed pulmonary edema, methemoglobinemia, deep burns, or systemic toxicity.

FIELD PRIORITIES

- Treat as hazardous until identified. Brush dry material before irrigation unless authoritative guidance directs otherwise.
- For most liquid eye/skin exposures, remove contaminated items and irrigate with copious tepid water.
- Do not induce vomiting, give oral neutralizers, or attempt acid-base neutralization. Contact Poison Help and HazMat for agent-specific direction.

CRITICAL ALERT

Water-reactive exception: do not apply water to a known or suspected water-reactive material until HazMat or Poison Help confirms the decontamination method.

App cross-links: TE-07 Overdose / Toxic Ingestion for opioids and naloxone; TE-02 Carbon Monoxide / Cyanide; TE-08 WMD / Nerve Agent; TB-02 Chemical / Electrical Burn.

DRAFT FOR REVIEW. Specialty therapies, local PPE/SCBA limits, decontamination resources, responder-exposure procedures, and receiving-facility notifications require medical-director, HazMat, pharmacy/formulary, and agency approval.

References: Tennessee EMS BLS/ALS State Protocol Guidelines 2024-2025, Hazardous Materials SOPs 801-809; HRSA Poison Help; U.S. DOT Emergency Response Guidebook; CDC/ATSDR Medical Management Guidelines; Claiborne County EMS local policies.